# Material cases: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

## E01 Diffusion scale

Convert $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$. From $\langle x^2 \rangle = 2Dt$, $t = (2 \times 10^{-6})^2/(2 \times 10^{-21}) = 2 \times 10^{9}\,\mathrm{s} \approx 63.4$ years. This is the chosen free-diffusion scale, not a SET measurement: electric-field drift, interfaces and film transport are excluded.

## E02 Parallel RC

$RC = (10^6)(10 \times 10^{-12}) = 10^{-5}\,\mathrm{s} = 10\,\mu\mathrm{s}$. Since $Z = R/(1 + j\omega RC)$, the maximum of $-\operatorname{Im} Z$ occurs at $\omega RC = 1$, so $f = 1/(2\pi RC) = 15.9\,\mathrm{kHz}$. No: distinct physical structures may have a similar equivalent impedance.

## E03 Weibull probability and median

$F(0.8) = 1 - e^{-(0.8)^4} = 1 - e^{-0.4096} = 0.3361$. With $F(x_{0.5}) = 0.5$, $x_{0.5} = 1[-\ln(0.5)]^{1/4} = 0.9124\,\mathrm{V}$. The result is conditional on the assumed model and its parameters.

## E04 Weibull quantiles and data

$x_p = \eta[-\ln(1-p)]^{1/4}$ gives $x_{0.1} = 0.5697\,\mathrm{V}$ and $x_{0.9} = 1.2318\,\mathrm{V}$. Successive cycles share device geometry and history, so they can be correlated; report device count separately, together with censoring, selection, binning and uncertainty as appropriate.

## E05 Heating and scaling

$P = V^2/R = 0.81/150 = 0.0054\,\mathrm{W} = 5.4\,\mathrm{mW}$ and $\Delta T = R_{\mathrm{th}} P = 162\,\mathrm{K}$. If RESET is modeled by constant $\Delta T_c$, then $V_{\mathrm{RESET}}^2 = \Delta T_c R/R_{\mathrm{th}}$. Independence of $R$ additionally requires the ratio $R/R_{\mathrm{th}}$ to stay fixed; steady operation and the lumped model must also be appropriate.

## E06 $HfO_2$/Ti aging evidence

The ratio is smaller by $10^4/10^2 = 100$. One ratio per time does not separately identify either absolute resistance or an oxygen trajectory. Depth-profile XPS may support a change in elemental or chemical distribution, subject to depth resolution, sputter effects and sample differences; it is not direct real-time tracking of an operating device.

# Material cases: Worked solutions (continued)

## E07 Classical nucleation

Differentiation gives $r^* = 2\gamma/|\Delta g_v| = 0.06/(3\times10^7) = 2.0\times10^{-9}\,\mathrm{m} = 2.0\,\mathrm{nm}$. Substitution yields $\Delta G^* = f16\pi\gamma^3/(3|\Delta g_v|^2) = 5.03\times10^{-20}\,\mathrm{J} \approx 0.314\,\mathrm{eV}$. The geometry and constant material parameters are model assumptions.

## E08 Thermal factor and rate

$\Delta G^*/k_BT \approx 0.314/0.02585 = 12.15$, so the factor is approximately $5.3\times10^{-6}$. It is dimensionless. A rate also requires a kinetic prefactor and a number of available sites; atomistic cluster size, interface geometry and nonequilibrium transport can invalidate the classical approximation.

## E09 Fast pulse

$I_C = C\,dV/dt = 0.2\times10^{-12}\times10^9 = 2\times10^{-4}\,\mathrm{A} = 0.2\,\mathrm{mA}$. With the specified definitions the SET time is $1.2\,\mathrm{ns}$, not the time of the earlier capacitive current. Energy depends on the full time-dependent voltage and device current, $E = \int V(t)I_{\mathrm{device}}(t)dt$, and on the chosen interval and parasitic-energy treatment; a peak does not supply that integral.

## E10 Three-terminal switch

A two-terminal device bridges its two electrodes. In the three-terminal example, the gate supplies or regulates ions; a separately biased source–drain pair probes and can be bridged by a conductive path. A path to the gate is a different, unwanted short. Ion migration and redox need not follow a MOSFET's field-effect channel mechanism, even though both devices have three terminals.

## E11 CBRAM reactions and humidity

Illustrative half-reactions are $\mathrm{Cu} \rightarrow \mathrm{Cu}^+ + e^-$ and $\mathrm{Cu}^+ + e^- \rightarrow \mathrm{Cu}$. Growth depends on transport, nucleation and local electrochemical reactions, so polarity alone is insufficient. A changing ion concentration and approximately unchanged estimated diffusivity do not demonstrate increased mobility; the reported behavior is specific to the sample and conditions and cannot establish a universal humidity trend.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

# Material cases: Worked solutions (continued)

## E12 Evidence and alternatives

Hysteresis at the stated temperatures is the observation, not identification of a mechanism. Under matched pulse conditions, vary temperature or the thermal boundary and compare timing and energy; a systematic trend can support a thermally assisted process but is not unique evidence. Compare synchronized electrical cycling with composition-sensitive imaging and matched controls: correlation can support metal rearrangement but probe perturbation and spatial resolution remain limiting. Measure a dummy structure or vary edge rate and bandwidth to estimate capacitance-driven transients; this can reveal parasitic contributions but does not on its own identify the switching material. Multiple independent controls are needed to distinguish competing models.

## E13 Conditional field drift

$\mu = D/(k_B T/e) = 10^{-21}/0.02585 = 3.868 \times 10^{-20}\,\mathrm{cm^2/(V\,s)}$. Thus $v_d = \mu E = 3.868 \times 10^{-15}\,\mathrm{cm/s}$ and, with $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$, $L/v_d = 5.17 \times 10^{8}\,\mathrm{s}$. This deliberately idealized scale omits diffusion, contact reactions and a varying field; linear mobility and Einstein's relation are assumed, and the field value alone does not establish them. In a hopping model, $qaE \ll k_B T$ is one condition to check. This is not a measured SET time.

## E14 RESET current bins

Using $I = 0.9/R$, the currents are respectively $9$, $6$, $4.5$, $3.6$, $3$, and $2.25\,\mathrm{mA}$. In the steady model $\Delta T = R_{\mathrm{th}} V^2/R$, the same critical temperature rise produces $|V| = \sqrt{\Delta T_c R/R_{\mathrm{th}}}$. A constant peak voltage therefore requires a common $\Delta T_c$ and approximately constant $R_{\mathrm{th}}/R$, as well as an adequate lumped, steady model. The six values are assumptions, not experimental data.

## E15 Ti-thickness comparison

$(5 \times 10^4)/10 = 5 \times 10^3$: the first ratio is 5,000 times the last. These are group-level, condition-specific ratios; they do not independently determine HRS and LRS or isolate Ti thickness from fabrication, sampling and storage. Different device batches, read polarities and added interlayers cannot be silently treated as the same experiment. XPS profiling adds support for redistribution but can be affected by preparation and sputtering.

# Material cases: Worked solutions (continued)

## E16 Competing timing laws

At 1, 1.5 and 2 V, $\ln(t/\mathrm{ns}) = 4, 3, 2$, giving $e^4 = 54.6$, $e^3 = 20.1$, and $e^2 = 7.39\,\mathrm{ns}$. The condition $\Delta G_0 > 0.10\,\mathrm{eV}$ keeps the assumed nucleation barrier positive over this range. For the assumed migration law, $\ln t = \ln(L/v_0) - [a/(2L(k_BT/e))]V$, and the magnitude of the slope is $(0.3/3)/(2 \times 0.025) = 2\,\mathrm{V}^{-1}$. Equal slopes do not select a unique mechanism; vary temperature, gap thickness and other controlled conditions and test independent composition or timing evidence.

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